

SHUFAN SHAHI

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EDUCATION

Islamic University of Technology 2022–2026
B.Sc. in Computer Science and Engineering

- Current CGPA: 3.55
- Relevant coursework: Data Structures and Algorithms, Machine Learning, Deep Learning with CV, Database Systems, and Computer Networks

EXPERIENCE

Student Researcher 2026–Present
Elite Research Lab

- Support research projects through literature reviews, experiments, and analysis.

Researcher 2025–2026
IUT Computer Vision Lab

- Researched multimodal AI methods to improve integration and mitigate modality imbalance.

Senior Executive 2025–2026
Project Aero

- Contributed to drone R&D by applying AI-based object tracking for international UAV competitions.

PUBLICATIONS AND CONFERENCES

EthMuSAM : Mutual Scoring With SAM For Zero Shot Anomaly Segmentation[Preprint] 2026

- Extracted patch-level representations from images using pretrained DINOv2 encoders.
- Localized anomalous regions by comparing test-image embeddings with representations from normal reference images.

Prototype-Guided Adaptive Feature Learning for Multimodal Emotion Recognition in Conversation (MERC)[Preprint] 2026

- Developed graph-compatible adaptive feature weighting and modality-specific prototype supervision to learn class-discriminative representations before cross-modal reasoning.
- Improved multimodal emotion-recognition performance across multiple benchmark datasets through adaptive feature processing and prototype-guided learning.

Rethinking Visual Features in Graph Based Multimodal Emotion Recognition in Conversations (MERC): From Low Level Pixels to Semantic Representations[Preprint] 2026

- Developed a semantic visual-feature pipeline that converts utterance-level videos into temporally structured descriptions of facial expressions, gaze, body language, and emotional transitions.
- Encoded the descriptions into dense representations for graph-based multimodal reasoning, improving cross-modal alignment while providing a viable alternative to noisy handcrafted visual features.

PROJECTS

Drone Vision - Aerial Tracking | *PyTorch, YOLOv11X, ByteTrack*

 [GitHub](#)

- Fine-tuned YOLOv11X on VisDrone and integrated ByteTrack for human counting and multi-object tracking in aerial footage.

BD Traffic Monitoring System - Traffic Analytics | *PyTorch, YOLO11L, Flask, Docker*

 [GitHub](#)

 [Live](#)

- Fine-tuned YOLO11L on Bangladesh traffic data to detect local vehicle classes and deployed the inference application with Flask and Docker.

Biztrack - Business Analytics | *Next.js, Node.js, PostgreSQL*

 [GitHub](#)

 [Demo](#)

- Built AI modules for real-time financial tracking, risk alerts, customer segmentation, and forecasting.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java

Data and Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

Databases: MySQL, Firebase, PostgreSQL

Web Development: React, Tailwind CSS, Next.js, Spring Boot

Mobile Development: React Native, Flutter

Visualization: Matplotlib, Seaborn

Development Tools: Git, Linux

ACHIEVEMENTS, AWARDS AND SCHOLARSHIPS

- **5th Place, VAND4 International Competition** 2026
Zero-Shot Industrial Track; built a zero-shot computer vision pipeline for detecting anomalies in industrial images using foundation models.
- **First Runner-up, IUT Coderush Hackathon** 2025
Built an application under tight deadlines in a fast-paced team.
- **7th Place, BAIC Datathon** 2025
Analyzed datasets using preprocessing, visualization, and modeling.
- **IUT Partial Scholarship** 2022
Secured a partial scholarship covering all four years of B.Sc. studies.
- **Runner-up, State-Level Physics Olympiad** 2018

REFERENCE

Shahriar Ivan

Assistant Professor, Department of CSE, IUT

Specialization: Machine Learning, Deep Learning, Computer Vision

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